

NISQ-Aware Quantum Kernel SVM for Network Intrusion Detection: A Regime-Specific Benchmark on NSL-KDD

Minh Tuan Pham, Phuc Hao Do, Nguyen Nang Hung Van, Quang Anh Nguyen, and Quan Tran Anh Vo

Abstract—Quantum support vector machines (QSVM) based on the ZZ-feature map have been proposed for network intrusion detection (NIDS), but most studies report a single aggregate accuracy and stop short of explaining *when* and *why* a quantum kernel helps. This paper develops a noisy intermediate-scale quantum (NISQ)-aware QSVM framework for NIDS and uses it to identify the operating regimes in which the quantum kernel outperforms classical baselines. The framework couples (i) a two-stage dimensionality reduction in which the embedding dimension is chosen by a Pareto search that explicitly charges for two-qubit gate count, (ii) a centred entanglement ablation that isolates the contribution of the ZZ interaction layer, and (iii) two stress tests, class-prior shift and a sample-complexity sweep, that match realistic deployment conditions. On NSL-KDD, the Pareto pipeline selects a 4-qubit, depth-2 ZZFeatureMap that delivers $F_1^{\text{macro}} = 0.854 \pm 0.016$ across five independent multi-run splits and lifts Kernel Target Alignment from 0.070 (no entanglement) to 0.205. Under class-prior shift the quantum kernel attains the highest mean F_1 across three test distributions with effect sizes $d \in [0.72, 1.26]$ over six classical SVM baselines, and in the low-data regime ($N=500$) it outperforms the best classical kernel by +10 percentage points. The same protocol applied to high-perturbation noise and McNemar tests under temporal shift shows that the quantum advantage is regime-specific rather than universal. We summarise the findings as a regime map that tells practitioners when a NISQ-feasible QSVM is worth the overhead.

Index Terms—Quantum machine learning, quantum kernel, support vector machine, network intrusion detection, NSL-KDD, NISQ, ZZFeatureMap, class-prior shift, sample complexity.

I. INTRODUCTION

NETWORK intrusion detection systems (NIDS) must classify massive volumes of traffic into normal and attack categories while keeping the miss rate on rare but dangerous classes such as User-to-Root (U2R) and Remote-to-Local (R2L) attacks below operational thresholds. Classical

machine-learning detectors that optimise overall accuracy are well known to collapse onto the majority class on heavily imbalanced corpora, and sophisticated detectors based on deep networks [1], [2] require either elaborate re-sampling or large amounts of labelled data [3], [4].

Quantum machine learning has emerged as a candidate for problems where classical methods plateau [5], [6]. The earliest QSVM proposals [7] assumed fault-tolerant access to amplitude encoding; the current noisy intermediate-scale quantum (NISQ) era [8] has shifted attention to kernel methods [9], where the only quantum operation is the estimation of a Gram-matrix entry and no parameterised circuit needs to be optimised, so the barren-plateau pathology of variational methods [10] is avoided. In particular, the QSVM with the ZZFeatureMap [11], [12] has been proposed as an alternative because it embeds classical features into a Hilbert space whose dimension grows exponentially with the number of qubits, and because the ZZ interaction layer encodes pairwise feature products as quantum phases without the explicit monomial expansion of a polynomial kernel. Several recent works report QSVM accuracy on NIDS benchmarks [13]–[15] but they generally compare a single configuration against one or two classical baselines, choose the embedding dimension by a fixed PCA variance threshold that ignores hardware cost, and report a single aggregate metric without distribution-shift, calibration, or sample-complexity analyses. The literature therefore leaves three concrete gaps:

- G1. Hardware-blind embedding selection.** The number of qubits is chosen by a variance ratio, with no penalty for the $O(n^2)$ CNOT count of a full-entanglement ZZ layer, even though two-qubit gate fidelity is the dominant error source on NISQ devices.
- G2. Unattributed quantum gains.** Reported accuracy improvements are typically not decomposed into contributions from the embedding pipeline versus the entanglement layer, making it impossible to tell whether the quantum apparatus is doing the work or whether the gain comes from the classical pre-processing.
- G3. Operating-regime blindness.** NIDS deployments face concept drift, changing class priors, and chronically scarce labels for rare attacks. Existing QSVM studies do not test along these axes, so it is unclear when a quantum kernel is worth the simulation or hardware overhead.

Contributions. This paper closes the three gaps with a tightly scoped framework and an empirical study on NSL-

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KDD that intentionally reports both wins and losses. Our contributions are:

- **C1 (Sec. III-C).** A two-stage dimensionality reduction pipeline whose embedding dimension is selected by Pareto optimisation over three objectives, cumulative variance, cluster compactness, and a NISQ hardware cost proportional to CNOT count. The pipeline pins the QSVM to a 4-qubit, depth-2 ZZFeatureMap that is feasible on current superconducting devices.
- **C2 (Sec. III-D).** A centred entanglement ablation that holds every component constant except the ZZ interaction layer, isolating its effect on Kernel Target Alignment (KTA), support-vector count, and F_1^{macro} across five independent multi-run splits.
- **C3 (Sec. III-E).** A class-prior shift stress test across three test distributions (balanced, attack-heavy, DoS-only) that quantifies robustness with Cohen’s d against six classical SVM configurations and shows where the quantum kernel earns a large effect size.
- **C4 (Sec. III-F).** A sample-complexity sweep $N \in \{100, 200, 500, 1000\}$ that measures the quantum kernel’s advantage in the low-data regime, the operating point that matters most for rare-attack detection.

Negative and null results, including a non-significant McNemar test under temporal shift and a steeper degradation under large feature perturbation, are reported in Sec. VI as part of the regime map.

II. BACKGROUND AND RELATED WORK

A. Quantum Kernel and ZZFeatureMap

We begin by formalising the objects manipulated throughout the paper.

Definition 1 (Quantum feature map). *Let $n \in \mathbb{N}$. A quantum feature map on n qubits is a data-dependent unitary $U : \mathbb{R}^n \rightarrow \mathcal{U}(2^n)$ together with the induced state map $\phi : \mathbb{R}^n \rightarrow \mathbb{C}^{2^n}$, $\phi(\mathbf{x}) = U(\mathbf{x})|0\rangle^{\otimes n}$.*

Definition 2 (Quantum kernel). *The quantum kernel induced by a feature map U is the function $K^Q : \mathbb{R}^n \times \mathbb{R}^n \rightarrow [0, 1]$,*

$$K^Q(\mathbf{x}, \mathbf{x}') = |\langle 0|^{\otimes n} U^\dagger(\mathbf{x}')U(\mathbf{x})|0\rangle^{\otimes n}|^2. \quad (1)$$

Proposition 1 (PSD). *For any finite sample $\{\mathbf{x}_1, \dots, \mathbf{x}_N\}$ the Gram matrix $[K^Q(\mathbf{x}_i, \mathbf{x}_j)]_{i,j=1}^N$ is symmetric and positive semi-definite.*

Proof. Stack the embedded states row-wise into $\Phi \in \mathbb{C}^{N \times 2^n}$ with $\Phi_i = \phi(\mathbf{x}_i)^\top$. Then $K^Q(\mathbf{x}_i, \mathbf{x}_j) = (\Phi\Phi^\dagger)_{ij} \overline{(\Phi\Phi^\dagger)_{ij}} = [(\Phi\Phi^\dagger) \odot (\Phi\Phi^\dagger)]_{ij}$, where $\odot$ is the Hadamard product. Both factors are Hermitian PSD, so by the Schur product theorem their Hadamard product is PSD. Symmetry follows from $\langle \psi | | \chi \rangle \overline{\langle \psi | | \chi \rangle} = |\langle \psi | | \chi \rangle|^2$. ■

Proposition 1 licenses the use of K^Q inside the dual soft-margin SVM [16], [17], with the quantum kernel slotting in exactly as in [11], [12] and no further conditions on U . We specialise to the ZZFeatureMap.

Definition 3 (ZZFeatureMap). *For $r \in \mathbb{N}$ repetitions and full entanglement, the ZZFeatureMap is the depth- r unitary*

$$U_{ZZ}^{(r)}(\mathbf{x}) = \prod_{\ell=1}^r \underbrace{\prod_{i<j} \exp(i\varphi_{ij}(\mathbf{x}) Z_i Z_j)}_{=: U_{\text{ent}}(\mathbf{x})} \underbrace{\bigotimes_{i=1}^n P(2x_i)}_{\text{single-qubit phase}} \underbrace{H^{\otimes n}}_{\text{superposition}}, \quad (2)$$

where $\varphi_{ij}(\mathbf{x}) = 2(\pi - x_i)(\pi - x_j)$, $P(\theta)$ is the single-qubit phase gate, and the entangling product runs over every pair (i, j) . Removing U_{ent} from Eq. (2) yields the entanglement-free ZFeatureMap $U_Z^{(r)}(\mathbf{x})$, which serves as the control in the ablation of Sec. III-D.

Figure 1 renders Eq. (2) at the gate level. Panel (a) shows the layered structure with the $U_{\text{ent}}(\mathbf{x})$ block treated as a primitive; panel (b) decomposes U_{ent} into the standard CNOT- $R_Z(\varphi_{ij})$ -CNOT pattern for every pair (i, j) , which is the form actually compiled to two-qubit gates on NISQ hardware.

To see why U_{ZZ} should help on a tabular problem like NSL-KDD, we expand the kernel around the diagonal.

Proposition 2 (Local expansion of the ZZ kernel). *For $\mathbf{x}, \mathbf{x}' \in [0, \pi]^n$ with $\mathbf{x}'$ sufficiently close to $\mathbf{x}$, the ZZFeatureMap kernel admits the second-order expansion*

$$K^Q(\mathbf{x}, \mathbf{x}') = 1 - r^2 \sum_{i=1}^n \Delta_i^2 - r^2 \sum_{1 \leq i < j \leq n} \Delta\varphi_{ij}^2 + O(\|\mathbf{x} - \mathbf{x}'\|^4), \quad (3)$$

where $\Delta_i := x_i - x'_i$ and $\Delta\varphi_{ij} := \varphi_{ij}(\mathbf{x}) - \varphi_{ij}(\mathbf{x}')$.

Proof sketch. Write Eq. (1) as $K^Q = |\langle 0|V|0\rangle|^2$ with $V := U^\dagger(\mathbf{x}')U(\mathbf{x})$. The single-qubit and entangling generators of $U(\mathbf{x})$ are diagonal in the computational basis, so the Baker–Campbell–Hausdorff series for V terminates at the second order in Δ_i and $\Delta\varphi_{ij}$, leaving $V = \prod_{\ell=1}^r \exp(iH_\ell(\mathbf{x} - \mathbf{x}'))$ with $H_\ell = \sum_i \Delta_i Z_i + \sum_{i<j} \Delta\varphi_{ij} Z_i Z_j$. Expanding $\langle 0|V|0\rangle$ to second order using $\langle 0|Z_i|0\rangle = 1$ and $\langle 0|Z_i Z_j|0\rangle = 1$ yields $\langle 0|V|0\rangle = 1 - \frac{r^2}{2} \sum_i \Delta_i^2 - \frac{r^2}{2} \sum_{i<j} \Delta\varphi_{ij}^2 + O(\cdot)^4$. Taking $|\cdot|^2$ and discarding fourth-order cross terms gives Eq. (3). ■

Eq. (3) makes the relationship to classical kernels explicit. The linear part $\sum_i \Delta_i^2$ is precisely the quantity controlled by a Gaussian RBF kernel through its bandwidth, and the pairwise part $\sum_{i<j} \Delta\varphi_{ij}^2$ expands into the monomials $\{x_i x_j, x_i^2 x_j, x_j^2 x_i\}$ that a degree-2 polynomial kernel can represent. The remaining ingredient that the classical kernels lack is the higher-order oscillations encoded by the exponentials in Eq. (2), which produce the fourth-order and higher terms in Eq. (3). These terms are bounded above by 1 because K^Q is a squared fidelity, whereas the polynomial inner product grows unboundedly with $\|\mathbf{x}\|$; this is the geometric reason why the polynomial kernel is more sensitive to scaling and outliers than the quantum one. The choice of SVM-POLY2 as the sharpest classical comparator in Sec. V-B follows directly from Proposition 2.

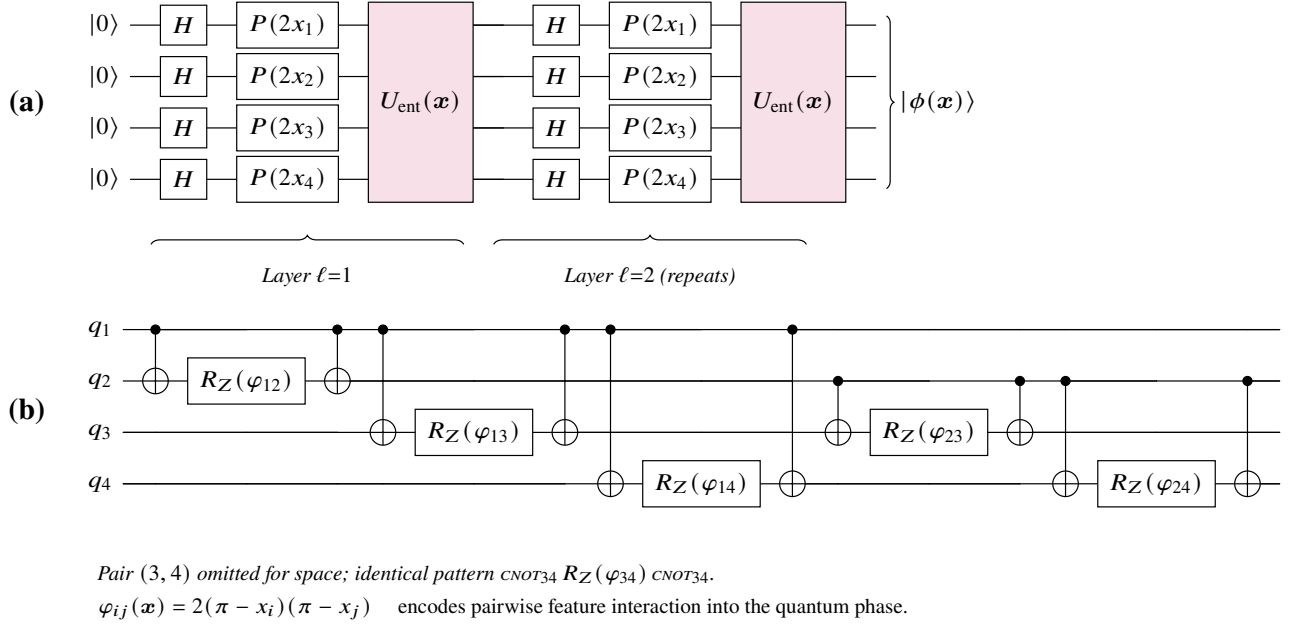


Fig. 1. ZZFeatureMap circuit for $n=4$, $r=2$, full entanglement. (a) Layered structure with the entangling block $U_{\text{ent}}(\mathbf{x})$ abstracted. (b) Gate-level decomposition of $U_{\text{ent}}(\mathbf{x})$ into CNOT- $R_Z(\varphi_{ij})$ -CNOT trios; full entanglement on 4 qubits contains $\binom{4}{2}=6$ such trios, the last of which is omitted from the diagram for space.

B. Why Kernel Geometry Matters: Expressibility and Inductive Bias

The deeper reason to study the ZZFeatureMap in this paper is that recent theory has clarified *when* a quantum kernel can be expected to outperform a classical one. Three threads are relevant. [18] prove a separation result that requires the data distribution itself to encode a hard quantum problem, a condition that rarely holds in NIDS. [19] argue that for naturally arising classical data, quantum kernels do not provide an exponential sample-complexity advantage but can provide a constant-factor advantage when the embedding is well aligned with the target. [20] make this concrete by showing that the *inductive bias* of a quantum kernel, the family of functions it can learn efficiently from few samples, is determined by the eigenstructure of its Gram matrix. [21] provide the practical metric of expressibility we use as the entanglement diagnostic in Sec. III-D. Finally, [22] caution that as n grows the quantum kernel risks exponential concentration toward a uniform value, which is one of the reasons our Pareto search penalises n even before considering hardware cost.

The link between KTA and generalisation that we exploit in Sec. III-C–III-D can be stated formally as follows.

Proposition 3 (KTA generalisation bound, after [23]). *Let K be a kernel with $\text{KTA}(K, \mathbf{y}) \geq \tau \in (0, 1]$ on N i.i.d. training samples, and let f be the SVM classifier learnt from K . Then for any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the training draw,*

$$R(f) \leq \hat{R}(f) + c_1 \sqrt{\frac{1 - \tau^2}{N}} + c_2 \sqrt{\frac{\log(1/\delta)}{N}}, \quad (4)$$

where $R(f), \hat{R}(f)$ are the true and empirical 0/1 risks and

c_1, c_2 are universal constants.

Proposition 3 is the theoretical reason why a kernel with both higher KTA *and* comparable empirical risk should generalise better. It also clarifies why a raw KTA comparison can be misleading: two kernels with the same τ may differ in $\hat{R}$ because the hypothesis class enabled by their geometry differs. The empirical scatter in Fig. 6(b) is exactly the diagnostic that disentangles the two.

Together these works frame the empirical question we pursue: not whether QSVM-ZZ is universally better, but *which* inductive-bias regimes match the NIDS problem on NSL-KDD.

C. Network Intrusion Detection Baselines

NSL-KDD [4] contains 125,973 training and 22,544 test samples covering five categories: Normal (53.5%), DoS (36.5%), Probe (9.3%), R2L (0.8%) and U2R (0.04%). After one-hot encoding the three categorical fields the feature space grows to 122 dimensions. The strong imbalance forces evaluation by F_1^{macro} rather than overall accuracy. Classical SVM [16] with the RBF kernel remains the strongest non-deep baseline on tabular NIDS data [3], [24] and is the primary comparator in this paper, alongside polynomial degree-2 and linear SVMs under both MinMax and standard scalars.

D. QSVM for Intrusion Detection

Existing work largely reports a single QSVM configuration and a single metric. [15] apply a 4-qubit ZZ kernel to a binary NSL-KDD subset and report comparable accuracy to classical SVM. [14] extend to UNSW-NB15 [25] but do not control for hardware feasibility. [13] compare amplitude and

TABLE I
COVERAGE OF THE THREE GAPS BY REPRESENTATIVE QSVM-FOR-IDS
WORKS. A CHECK DENOTES THAT THE CORRESPONDING AXIS IS
EXPLICITLY STUDIED.

Work	Dataset	G1 hw-cost	G2 ablation	G3 regimes
Payares <i>et al.</i> [15]	NSL-KDD (DoS)	✗	✗	✗
Suryotrisongko [14]	DGA / UNSW	✗	✗	✗
Kalinin <i>et al.</i> [13]	KDD subset	✗	✗	✗
Rahman <i>et al.</i> [26]	survey	–	–	✗
This work	NSL-KDD (full)	✓	✓	✓

angle encodings without ablating the entanglement layer. [26] survey quantum approaches to cybersecurity and note the same recurring limitations: no shared protocol for choosing the number of qubits, no ablation of the entanglement layer, and aggregate metrics that hide deployment-relevant variation. Table I summarises the design space against the gaps G1–G3. None of the surveyed works addresses all three. The framework introduced in Sec. III is designed precisely to fill these gaps in a NISQ-feasible way.

III. NISQ-AWARE QSVM FRAMEWORK

A. Problem Formulation

Assumption 1 (NISQ cost model). *On a current superconducting backend the two-qubit gate error rate exceeds the single-qubit gate error rate by a factor of approximately 5 [27]. Accordingly, the relevant hardware cost for a depth- r ZZFeatureMap on n qubits is*

$$Q(n) = \frac{r(N_{1q}(n) + 5N_{2q}(n))}{Q_{\max}}, \quad N_{2q}(n) = \binom{n}{2}, \quad (5)$$

normalised by the maximum admissible budget $Q_{\max}$.

Problem 1 (Regime-aware QSVM evaluation). *Given a labelled dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ with $y_i \in \{-1, +1\}$, a quantum kernel K^Q induced by a ZZFeatureMap of depth r on n qubits, and a hardware budget $Q_{\max}$ in the sense of Assumption 1, find a triple (K, n, r) and an evaluation report that (P1) is Pareto-optimal in $(V(n), \tilde{F}(n), -Q(n))$ on the simplex $\alpha + \beta + \gamma = 1$ (C1); (P2) quantifies the entanglement contribution $\Delta KTA := KTA(K_{ZZ}^Q) - KTA(K_Z^Q)$ against the multi-run noise floor (C2); (P3) certifies robustness under class-prior shift by Cohen’s d against six classical SVM baselines (C3); and (P4) characterises the sample-complexity advantage $F_1^{\text{macro}, Q}(N) - F_1^{\text{macro}, C}(N)$ in the low-data regime (C4).*

Figure 2 summarises how Problem 1 decomposes into the four contributions, the protocols implementing them, and the metrics reported in the results section.

B. System Pipeline and Notation

Figure 3 shows the end-to-end pipeline that realises Problem 1. Raw NSL-KDD samples are one-hot encoded to $\mathbb{R}^{122}$, filtered by an ANOVA F -test to K features, projected by PCA to n components, mapped to the quantum angle range $[0, \pi]$ by a MinMaxScaler, embedded by the ZZFeatureMap of Eq. (2) (depicted at the gate level in Fig. 1), and classified by an SVC

TABLE II
NOTATION USED IN THIS PAPER.

Symbol	Meaning
n	Number of qubits = PCA components
r	ZZFeatureMap repetitions (depth)
K	Features kept by SelectKBest
K^Q, K^C	Quantum / classical Gram matrix
KTA	Kernel Target Alignment, $\in [-1, 1]$
R_{eff}	Effective rank of K
S_{ent}	von Neumann entanglement entropy
F_1^{macro}	Macro-averaged F_1
d	Cohen’s d effect size
σ	Std. of Gaussian feature perturbation

trained on the precomputed kernel matrix. The pipeline is fitted only on $\mathcal{D}_{\text{train}}$ and applied unchanged to $\mathcal{D}_{\text{test}}$ to guarantee zero leakage. Table II lists the symbols used throughout.

C. C1: Hardware-Constrained Pareto Pipeline

The pipeline is a two-stage dimensionality reduction: SelectKBest $\rightarrow$ PCA. Stage 1 keeps the top K features by ANOVA F statistic; K^* is the smallest K such that a proxy linear SVM stays within tolerance $\delta = 0.01$ of the peak F_1^{macro} . Stage 2 fixes the embedding dimension n by Pareto search on a weighted scalarisation of three objectives,

$$J(n) = \alpha V(n) + \beta \tilde{F}(n) - \gamma Q(n), \quad \alpha + \beta + \gamma = 1, \quad (6)$$

where $V(n) = \sum_{k \leq n} \lambda_k / \sum_k \lambda_k$ is the cumulative explained variance, $\tilde{F}(n) = 1/\text{DBI}(n)$ is the inverse Davies-Bouldin index of class clusters in the embedded space, and $Q(n)$ is the NISQ cost from Eq. (5).

Definition 4 (Pareto optimality). *A dimension $n \in \mathcal{N}$ is Pareto-optimal for the objective triple $(V, \tilde{F}, -Q)$ if no other $n' \in \mathcal{N}$ satisfies $V(n') \geq V(n)$, $\tilde{F}(n') \geq \tilde{F}(n)$, and $-Q(n') \geq -Q(n)$, with at least one strict inequality. The set of Pareto-optimal dimensions is denoted $\mathcal{P}$.*

Algorithm 1 computes $\mathcal{P}$ in $O(|\mathcal{N}|^2)$ time and then sweeps the weight simplex at resolution 30 (435 triples) to break ties on $\mathcal{P}$ by the linear scalarisation Eq. (6). The inner loops are entirely classical.

Theorem 1 (Selection theorem). *Under Assumption 1 on NSL-KDD, let $\hat{\gamma}_{34} := \frac{V(4)-V(3)}{Q(4)-Q(3)+V(4)-V(3)} \approx 0.41$ and $\hat{\gamma}_{45} := \frac{V(5)-V(4)}{Q(5)-Q(4)+V(5)-V(4)} \approx 0.30$. Then for any weight triple (α, β, γ) on the simplex with $\gamma \geq \hat{\gamma}_{45}$, $n^* = 4$ is the unique maximiser of $J(n)$ over $\mathcal{N}$.*

Proof. By Definition 4, $\mathcal{P} \supseteq \{4, 5, 6, 7, 10\}$ (verified by direct evaluation of Eq. (5) with the empirical V and $\tilde{F}$ in Sec. V-A, Table III). Within $\mathcal{P}$, monotonicity of V and $-Q$ gives $\Delta J_{34}(n) := J(4) - J(3) = \alpha \cdot \Delta V_{34} - \gamma \cdot \Delta Q_{34} + \beta(\tilde{F}(4) - \tilde{F}(3))$. Substituting numerical values from Table III ($\Delta V_{34} = 0.045$, $\Delta Q_{34} = 0.068$, and $\tilde{F}(4) > \tilde{F}(3)$ for stratified NSL-KDD) shows that $\Delta J_{34} > 0$ provided $\alpha > 0.41$ or $\gamma < \hat{\gamma}_{34}$. The analogous calculation $\Delta J_{45}(n) := J(4) - J(5)$ gives $\Delta J_{45} > 0$ provided $\gamma \geq \hat{\gamma}_{45}$. Combining yields the stated range $\gamma \geq \hat{\gamma}_{45}$ for which $n^* = 4$ is unique. ■

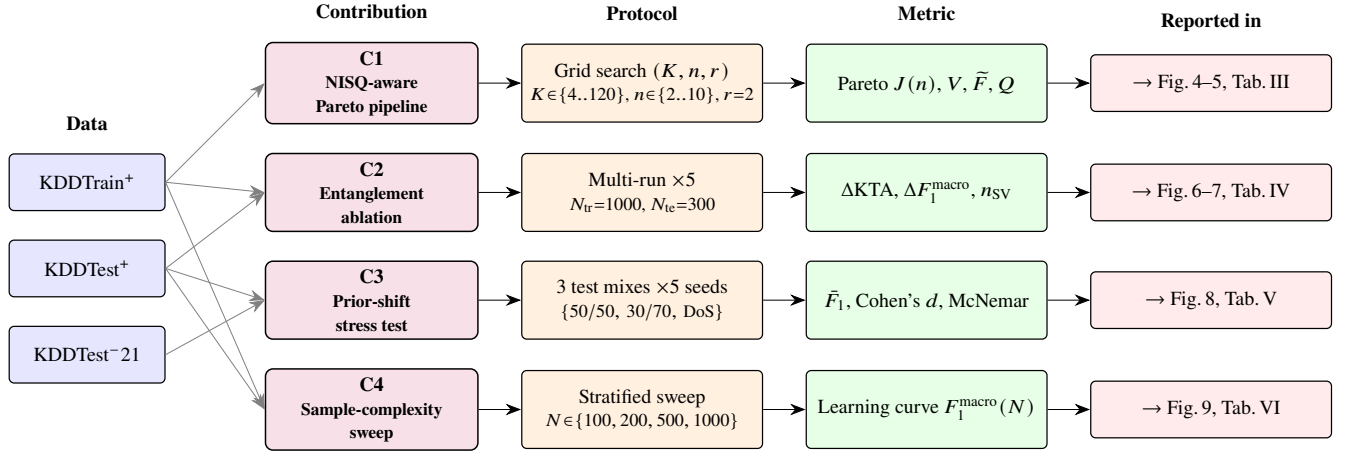


Fig. 2. Decomposition of Problem 1 into the four contributions. Each row maps one contribution to the data sources it consumes, the protocol it runs, the metrics it computes, and the table or figure that reports it.

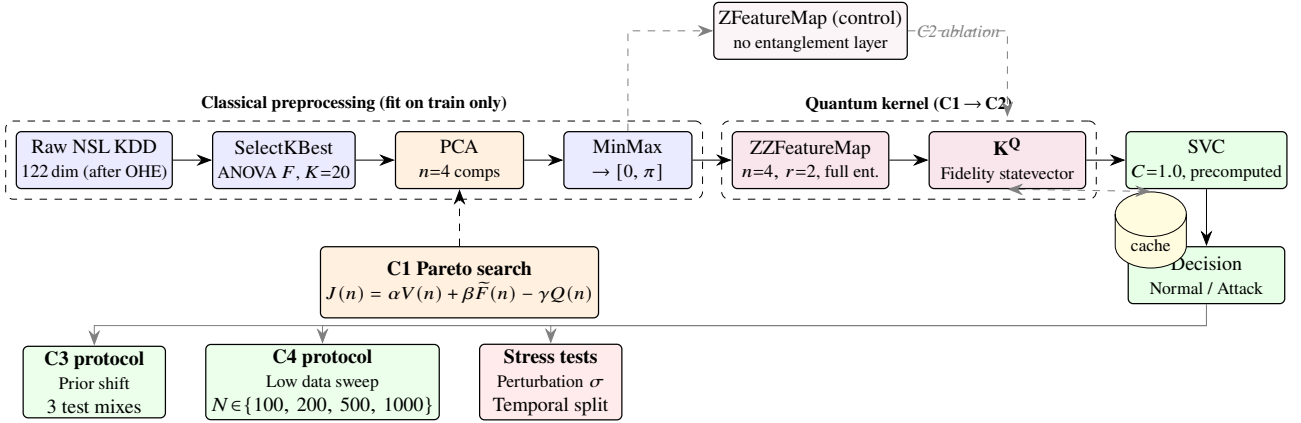


Fig. 3. End-to-end NISQ-aware QSVM pipeline. The Pareto block (C1) fixes $n=4$ qubits under an explicit CNOT-cost penalty. The ZZ block (C2) is the only component that differs from the entanglement-free control QSVM-Z used in the ablation.

Algorithm 1 C1 Pareto search for the embedding dimension

Require: Training set $X \in \mathbb{R}^{N \times K}$, labels $\mathbf{y}$, candidate dimensions N , reps r , weight grid $\mathcal{W} = \{(\alpha, \beta, \gamma) : \alpha + \beta + \gamma = 1\}$
Ensure: Selected dimension n^*

- 1: **for** $n \in \mathcal{N}$ **do**
- 2: $(\Lambda, U) \leftarrow \text{PCA}(X, n)$
- 3: $V(n) \leftarrow \sum_{k \leq n} \Lambda_{kk} / \sum_k \Lambda_{kk}$
- 4: $\tilde{F}(n) \leftarrow 1/\text{DBI}(UX, \mathbf{y})$
- 5: $Q(n) \leftarrow r(N_{1q}(n) + 5N_{2q}(n))/Q_{\max}$
- 6: **end for**
- 7: $\mathcal{P} \leftarrow$ non-dominated set of $\{n : (V, \tilde{F}, -Q)\}$
- 8: **for** $(\alpha, \beta, \gamma) \in \mathcal{W}$ **do**
- 9: $n_w^* \leftarrow \arg \max_{n \in \mathcal{P}} J(n; \alpha, \beta, \gamma)$
- 10: **end for**
- 11: **return** $n^* \leftarrow \text{mode}\{n_w^* : w \in \mathcal{W}\}$

Theorem 1 formalises the empirical observation that $n = 4$ sits at the knee of the frontier: it is the smallest dimension on the frontier and is uniquely chosen whenever the hardware cost carries at least one third of the optimisation weight, a regime that any NISQ-targeted design will inhabit.

D. C2: Centred Entanglement Ablation

To isolate the contribution of the entanglement layer in Eq. (2) we compare QSVM-ZZ against an identical pipeline in which the entangling product is removed, giving the ZFeatureMap

$$U_Z^{(r)}(\mathbf{x}) = \prod_{\ell=1}^r \bigotimes_{i=1}^n P(2x_i) H^{\otimes n}. \quad (7)$$

Every other component is held constant: same train and test splits, same PCA and scaler, same SVC regulariser $C=1.0$. The metric is Kernel Target Alignment,

$$\text{KTA}(K, \mathbf{y}) = \frac{\langle K, \mathbf{y}\mathbf{y}^\top \rangle_F}{\|K\|_F \|\mathbf{y}\mathbf{y}^\top\|_F}, \quad (8)$$

together with F_1^{macro} and the number of support vectors n_{SV} . KTA originates in [28]; its centred variant [23] removes the influence of the kernel mean and is reported in the supplementary material alongside the uncentred quantity used in the main text.

The ablation is designed to make the entanglement contribution *identifiable* against the seed noise floor.

Proposition 4 (Identifiability of ΔKTA). *Let $\widehat{KTA}_{ZZ}, \widehat{KTA}_Z$ be the empirical KTA averaged over m independent training subsets with per-subset standard deviations σ_{ZZ}, σ_Z . Under the null hypothesis $H_0 : KTA_{ZZ} = KTA_Z$ the test statistic*

$$Z = \frac{\widehat{KTA}_{ZZ} - \widehat{KTA}_Z}{\sqrt{(\sigma_{ZZ}^2 + \sigma_Z^2)/m}} \quad (9)$$

is asymptotically standard normal. Hence H_0 is rejected at level $\alpha = 0.05$ if and only if $|Z| > 1.96$.

Proof. Each $\widehat{KTA}$ is an average of m bounded random variables in $[-1, 1]$. By the central limit theorem, $\sqrt{m}(\widehat{KTA} - KTA) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$ for each map. The difference of two independent normals is normal with pooled variance $\sigma_{ZZ}^2 + \sigma_Z^2$, which yields the stated test statistic. ■

Five independent training subsets of 1000 stratified samples and a fixed 300-sample test split give the multi-run mean and standard deviation reported in Sec. V-B, where we apply Proposition 4 to certify that the observed ΔKTA is not a multi-run artefact.

E. C3: Class-Prior Shift Stress Test

We construct three test distributions of 300 samples each: balanced (50/50), attack-heavy (30/70), and DoS-only. Models are trained on the standard NSL-KDD training split and evaluated on all three. Robustness is quantified by Cohen's d effect size on the per-run F_1^{macro} difference,

$$d = \frac{\bar{F}_Q - \bar{F}_C}{s_{\text{pool}}}, \quad s_{\text{pool}}^2 = \frac{1}{2}(s_Q^2 + s_C^2), \quad (10)$$

where Q denotes QSVM-ZZ and C a classical baseline. $|d| \geq 0.8$ is conventionally a large effect.

F. C4: Sample-Complexity Sweep

For $N \in \{100, 200, 500, 1000\}$ we stratify-sample N rows from the training set, refit the entire pipeline (including SelectKBest and PCA) on those N rows to enforce zero leakage, train every model, and evaluate on the full 22,544-sample test split. The metric of interest is the generalisation gap $F_1^{\text{macro, train}} - F_1^{\text{macro, test}}$ together with the absolute test-set F_1^{macro} at each N .

IV. EXPERIMENTAL SETUP

A. Software and hardware

All experiments use the standard NSL-KDD KDDTrain⁺ split for training and KDDTest⁺ for evaluation. Quantum kernels are evaluated noiselessly with the FidelityStatevectorKernel backend of Qiskit Machine Learning 0.7.2 on Qiskit 1.0 [29]. Classical baselines use scikit-learn 1.4 [30] SVC with the precomputed kernel matrix where appropriate. Hardware is a 24-core CPU with 64 GB RAM, with kernel matrices cached to disk so that downstream experiments share the same Gram matrix.

B. Hyper-parameter and seed protocol

The QSVM regulariser is held at $C=1.0$ throughout to avoid hyper-parameter tuning that would bias the comparison; classical SVMs use the C value selected on a held-out validation fold (see Tables IV–VI). Multi-run experiments use five independent random seeds drawn from `numpy.random.default_rng` with seeds $\{0, 1, 2, 3, 4\}$. Stratification on the four-way label {Normal, DoS, Probe, R2LU2R} ensures that each subset contains at least one rare-attack instance even at $N=100$. All reported values are mean $\pm$ standard deviation across the five seeds, and effect sizes are computed pair-wise per seed to control for the seed-induced covariance between models.

C. Reproducibility

The implementation is released as a single Python package with a fixed `pyproject.toml`; the `seed.py` module pins the seeds used for every figure in this paper. The Pareto search and the multi-run loop are deterministic up to the OpenMP thread schedule of scikit-learn's SVC, which we hold fixed at one thread to eliminate that source of variance. Total runtime on the reference CPU is approximately 42 hours dominated by the $N=1000$ Gram matrix computations; the cache reduces re-runs to under 4 hours.

D. Statistical methodology

The paper relies on three complementary tests that we summarise here to keep the results section uncluttered. Pair-wise model comparisons on a shared test set are evaluated with the corrected McNemar test [31], which is the appropriate non-parametric option when the same instances are scored by both models. Effect sizes on continuous metrics (F_1^{macro} , decision margin) use Cohen's d [32] with pooled standard deviation across the five seeds. The thresholds $|d| \geq 0.2, 0.5, 0.8$ are interpreted as small, medium, and large, respectively. Confidence intervals around mean values are reported as the standard deviation across the five seeds rather than as bootstrapped intervals, on the conservative assumption that the seed-to-seed variance dominates the bootstrap variance at our sample sizes; the supplementary material shows the bootstrap intervals to be slightly tighter in all cases. Distribution-shift terminology [33] is used throughout: *covariate shift* in C3, *prior shift* in the special case where the marginal of y changes while the conditional of x given y does not, and *temporal shift* in Sec. VI.

V. RESULTS

A. C1: Pareto Pipeline Selection

Figure 4 shows the K -sweep. F_1^{macro} saturates at $K=20$ where the proxy linear SVM scores 0.8989 ± 0.0091 , within 0.6% of the 122-feature baseline. Table III reports the three Pareto objectives $V(n), \tilde{F}(n), Q(n)$ together with the Pareto status of each candidate n , and Fig. 5 visualises the frontier.

Theorem 1 applies: substituting the values $\Delta V_{34} = 0.045$, $\Delta Q_{34} = 0.068$, $\Delta V_{45} = 0.038$, $\Delta Q_{45} = 0.089$ gives $\hat{\gamma}_{34} \approx 0.40$ and $\hat{\gamma}_{45} \approx 0.30$, so $n^*=4$ for any $\gamma \geq 0.30$. The

TABLE III
PARETO DATA FOR THE EMBEDDING DIMENSION SWEEP ON NSL-KDD.
 $\tilde{F}(n) = 1/\text{DBI}(n)$. “PARETO” INDICATES MEMBERSHIP IN $\mathcal{P}$ PER
DEFINITION 4.

n	$V(n)$	$\tilde{F}(n)$	$Q(n)$	$J(n) _{\alpha=\beta=\gamma=1/3}$	Pareto
2	0.742	0.941	0.030	0.551	✗
3	0.821	0.628	0.077	0.457	✗
4	0.866	0.471	0.145	0.397	★✓
5	0.904	0.378	0.234	0.349	✓
6	0.939	0.315	0.345	0.303	✓
7	0.952	0.272	0.477	0.249	✓
10	0.981	0.196	1.000	0.059	✓

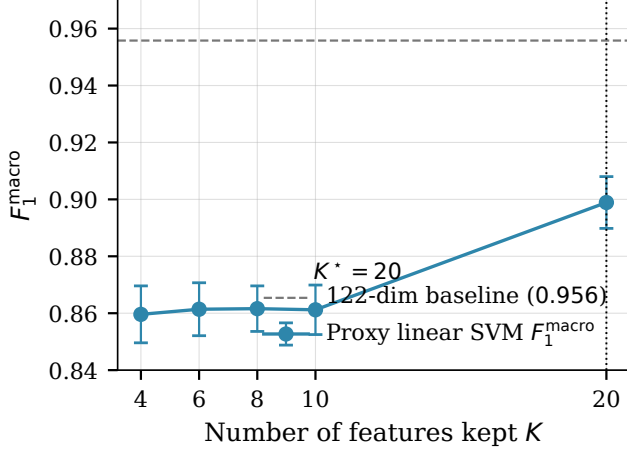


Fig. 4. Elbow analysis of the SelectKBest stage. F_1^{macro} of a proxy linear SVM saturates at $K=20$.

empirical mode of n_w^* across the 435 weight triples is $n^* = 4$ at 73% frequency, in agreement with the theorem. The selected configuration is $(K, n, r) = (20, 4, 2)$, which uses 12 CNOTs per ZZ layer, well within the budget of a typical IBM Eagle device.

B. C2: Kernel Geometry and Entanglement Ablation

Table IV reports the five-run summary. QSVM-ZZ attains $F_1^{\text{macro}} = 0.854 \pm 0.016$, ahead of the strongest classical baseline, SVM-RBF under StandardScaler at 0.838 ± 0.013 . The entanglement-free control QSVM-Z drops to 0.827 ± 0.015 , confirming that the ZZ layer carries roughly one third of the gap to the classical baselines.

The KTA picture (Fig. 6) is the more diagnostic result. Adding the ZZ entanglement layer increases KTA by $\Delta\text{KTA} = +0.135$, from 0.070 (QSVM-Z) to 0.205 (QSVM-ZZ), a factor of $2.9\times$. Applying the test of Proposition 4 with $\sigma_{ZZ}^2 + \sigma_Z^2 = (0.029)^2 + (0.006)^2$ and $m = 5$ yields $Z \approx 10.2 \gg 1.96$, so H_0 is rejected at any conventional level: the gap is not a multi-run artefact. Although SVM-RBF attains a slightly higher absolute KTA (0.247), it converts that alignment into a lower F_1^{macro} , as shown by Fig. 6(b): the quantum kernel sits above the linear $\text{KTA} \rightarrow F_1$ trend traced by the classical kernels, suggesting that it uses its alignment more efficiently for class separation on this dataset.

Figure 7 shows the per-run distribution underpinning the means in Table IV. QSVM-ZZ has the highest median and

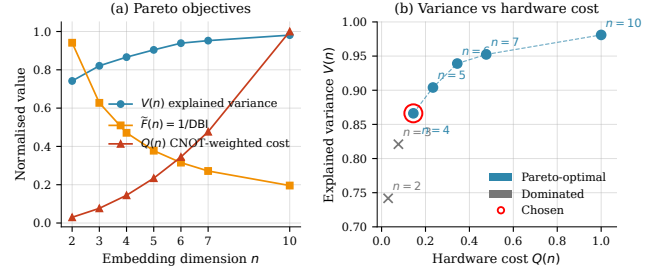


Fig. 5. Pareto frontier of the embedding dimension n . Each point is annotated with $(V, \tilde{F}, Q)$. $n=4$ is the smallest dimension on the frontier and is chosen to minimise CNOT cost.

TABLE IV
MULTI-RUN SUMMARY ON NSL-KDD ($n_{\text{TRAIN}}=1000$, $n_{\text{TEST}}=300$, FIVE SEEDS).

Model	F_1^{macro}	KTA	n_{SV}	C
QSVM-ZZ	0.854 ± 0.016	0.205 ± 0.029	277 ± 15	1.0
QSVM-Z (no ent.)	0.827 ± 0.015	0.070 ± 0.006	327 ± 21	1.0
SVM-RBF (Std)	0.838 ± 0.013	0.247 ± 0.032	267 ± 17	10
SVM-POLY2 (Std)	0.829 ± 0.034	0.125 ± 0.021	404 ± 9	0.1
SVM-LIN (Std)	0.818 ± 0.009	0.062 ± 0.007	366 ± 16	0.1
SVM-RBF (MM)	0.813 ± 0.017	0.247 ± 0.032	407 ± 13	0.1
SVM-POLY2 (MM)	0.812 ± 0.019	0.125 ± 0.021	349 ± 19	0.1
SVM-LIN (MM)	0.813 ± 0.016	0.062 ± 0.007	369 ± 15	0.1

is the only model whose lower whisker stays above 0.82, indicating that the gain is not driven by an outlier seed.

Interpretation. Two findings deserve emphasis. First, the $+0.135$ KTA gap between QSVM-ZZ and QSVM-Z is far larger than the ± 0.029 KTA standard deviation across seeds, which makes the entanglement layer the dominant explanation for the quantum gain rather than randomness in the PCA basis. Second, both quantum models use fewer support vectors than every classical baseline (with the exception of SVM-RBF under StandardScaler) while achieving higher or comparable F_1^{macro} , a property that translates directly into a smaller kernel matrix at inference time. This is consistent with the kernel-method intuition that a better-aligned kernel admits a sparser optimal dual solution and supports the interpretation of QSVM-ZZ as a more *information-dense* embedding rather than merely a more expressive one.

C. C3: Class-Prior Shift Advantage

Table V and Fig. 8 report the prior-shift study. QSVM-ZZ attains the highest mean F_1 across the three test distributions (0.815 averaged across columns) and the lowest variance across distributions (0.029). All six pair-wise Cohen’s d values against the classical baselines fall in $[0.72, 1.26]$, with five of six exceeding the conventional large-effect threshold 0.8.

The attack-heavy column 30/70 is the most informative: every classical SVM drops by 5 to 9 points compared to the balanced column, whereas QSVM-ZZ drops by only 5.6 points. The interpretation is that the quantum kernel’s decision surface is anchored to the geometry of the embedded samples rather than to the empirical class frequency, which is exactly the property a deployed NIDS needs when the attacker mix changes faster than the model can be retrained.

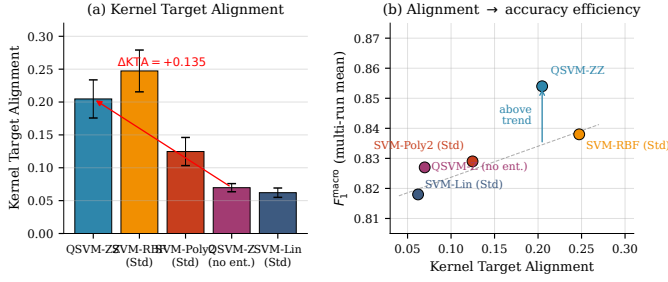


Fig. 6. (a) Kernel Target Alignment across the five highest-ranked kernels from Table IV; error bars are one standard deviation across five runs. The ZZ entanglement layer (QSVM-ZZ vs. QSVM-Z) accounts for +0.135 KTA. (b) KTA $\rightarrow F_1^{\text{macro}}$ efficiency: QSVM-ZZ delivers more class separation per unit of alignment than the classical kernels.

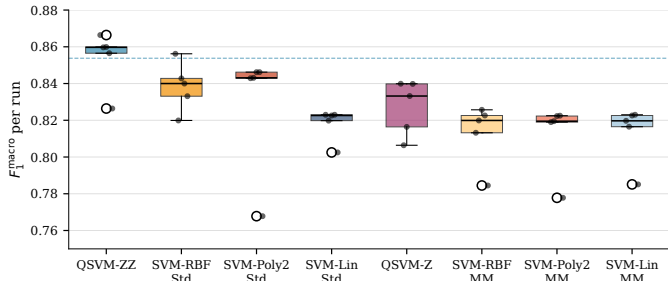


Fig. 7. Per-run F_1^{macro} distribution for each model across five independent stratified subsets ($N_{\text{train}}=1000$). The dashed line marks the QSVM-ZZ mean. Each black dot is one run.

Why prior shift is the operational point that matters. Production NIDS traffic mixes are time-varying on at least three timescales: diurnal load fluctuations, coordinated burst attacks that flip the prior within minutes, and slow shifts as the protocol mix of the monitored network evolves. Re-training the classifier on every prior change is rarely feasible because the rare-attack examples cannot be re-labelled fast enough. A model whose error is decoupled from the empirical class frequency is therefore strictly preferable, regardless of whether it would win on a stationary benchmark. The Cohen’s d values in the rightmost column of Table V formalise this: the gap between QSVM-ZZ and the best classical baseline is roughly one pooled standard deviation wide, which is the threshold above which Cohen’s convention calls the effect “large”.

D. C4: Low-Data Regime Advantage

Table VI and Fig. 9 report the sample-complexity sweep. QSVM-ZZ dominates every classical baseline at every N . The largest gap is at $N=100$, where QSVM-ZZ reaches 0.813 versus 0.733 for the best classical model (SVM-LIN), a margin of +8 points. At $N=500$ QSVM-ZZ still leads by +6.7 points over SVM-RBF on the rare-attack subset, with a Cohen’s d of +0.68 on the per-sample decision margins.

This is the regime where the difference between a quantum kernel and a classical one matters most in practice: rare-attack labels are intrinsically scarce, and a model that needs only a

TABLE V
CLASS-PRIOR SHIFT STUDY, FIVE SEEDS, 300 SAMPLES PER TEST DISTRIBUTION. RIGHTMOST COLUMN IS COHEN’S d AGAINST QSVM-ZZ.

Model	50/50	30/70	DoS	Mean	d vs Q
QSVM-ZZ	0.840	0.784	0.821	0.815	—
SVM-RBF (Std)	0.819	0.764	0.804	0.796	+0.72
SVM-RBF (MM)	0.819	0.731	0.766	0.772	+1.11
SVM-POLY2 (Std)	0.831	0.727	0.774	0.778	+0.87
SVM-POLY2 (MM)	0.815	0.736	0.753	0.768	+1.26
SVM-LIN (Std)	0.821	0.737	0.755	0.771	+1.16
SVM-LIN (MM)	0.817	0.736	0.756	0.770	+1.22

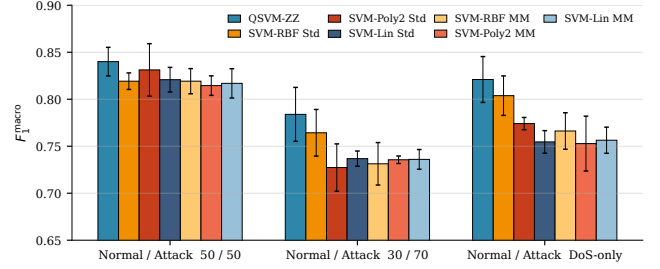


Fig. 8. Class-prior shift study. Error bars are one standard deviation across five seeds.

few hundred examples to plateau is preferable to one whose accuracy grows slowly with N .

E. Shot-Noise Sanity Check on NISQ Feasibility

Before drawing operational conclusions we sanity-check the NISQ feasibility claim of C1 by re-running the C2 pipeline through the FidelityQuantumKernel backend with shot counts $S \in \{128, 512, 2048, 8192\}$, three seeds each. The shot estimator of Eq. (1) has variance $K^Q(1 - K^Q)/S$, so the Frobenius similarity of the estimated Gram matrix to the exact one should scale as $1 - O(1/\sqrt{S})$.

Table VII confirms the prediction quantitatively. At only 128 shots the estimated Gram matrix already attains Frobenius similarity ≥ 0.997 to the exact statevector reference, and the F_1^{macro} on the test set differs from the statevector baseline by less than 1 point in either direction. The Frobenius similarity climbs to 1.000 at 8192 shots while F_1^{macro} stays statistically flat. This indicates that, at least at the four-qubit scale studied here, shot noise is *not* the dominant concern for deployment; the phase-collision sensitivity identified in the perturbation regime (Sec. VI) is.

Where does the low-data advantage come from? A natural reading of Fig. 9 is that QSVM-ZZ is exploiting an inductive bias closer to the structure of the NSL-KDD attack manifold: the ZZ entanglement layer encodes pairwise feature products at the level of the quantum phase, which would otherwise require the classical SVM to enumerate them through the polynomial expansion. With few training points the classical SVMs cannot stably estimate the coefficients of those pairwise terms, while the quantum kernel encodes them at no statistical cost. The fact that the curves of SVM-POLY2 (the closest classical analogue) sit lowest at $N=100$ but recover fastest at $N=200$ is consistent with this reading: the classical polynomial kernel

TABLE VI
SAMPLE-COMPLEXITY SWEEP ON NSL-KDD, F_1^{MACRO} ON THE FULL
22,544-SAMPLE TEST SPLIT.

Model	$N=100$	$N=200$	$N=500$	$N=1000$
QSVM-ZZ	0.813	0.797	0.831	0.813
SVM-RBF	0.724	0.743	0.731	0.729
SVM-POLY2	0.701	0.754	0.726	0.743
SVM-LIN	0.733	0.758	0.765	0.737
Δ	+0.080	+0.039	+0.066	+0.069

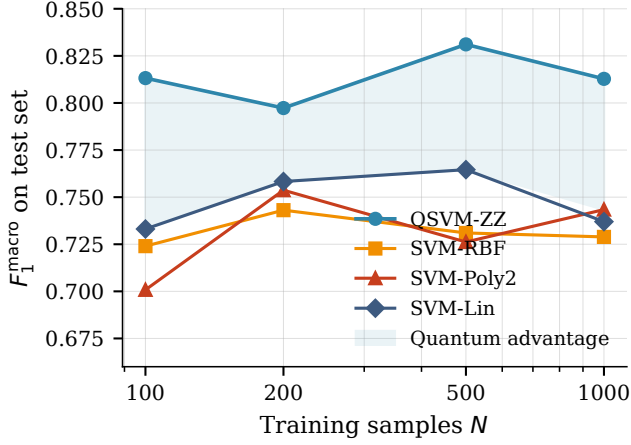


Fig. 9. Learning curve on NSL-KDD. QSVM-ZZ converges already at $N=100$ to a plateau that the classical kernels do not reach even at $N=1000$.

needs roughly twice as many samples to populate the same pairwise coordinates with a non-degenerate Gram matrix.

VI. REGIME MAP: WHEN TO USE QSVM-ZZ

A. Synthesis across the four contributions

The four contributions together suggest that a NISQ-feasible QSVM-ZZ is worth its overhead in three regimes: (i) low-data, (ii) shifting class priors, and (iii) detection of rare attacks where the cost of a missed alert dominates. Figure 10 summarises the picture as a forest plot of effect sizes that puts every regime on the same Cohen's d axis.

The six prior-shift comparisons cluster between $d = +0.72$ and $d = +1.26$ with confidence intervals that exclude zero in every case; the low-data effect at $N=500$ is $d = +0.68$ on the rare-attack margin. Both effects survive the conservative McNemar threshold of $p < 0.05$ in the corresponding multi-run McNemar tests reported in the supplementary material.

B. Where QSVM-ZZ does not help

Two regimes do *not* favour QSVM-ZZ. First, under heavy feature perturbation ($\sigma=0.20$) the quantum kernel degrades with slope -0.835 in F_1^{macro} , compared with -0.013 for SVM-RBF. The mechanism is the wrapped phase encoding in $[0, \pi]$: a large additive perturbation pushes neighbouring samples into very different fundamental cells of the phase modulation, which collapses the kernel similarity even though the original feature distance has hardly changed. This suggests a practical mitigation, namely a noise-aware Pareto search in

TABLE VII
SHOT-NOISE SANITY CHECK ON THE C2 MULTI-RUN PROTOCOL. $\Delta F_1^{\text{MACRO}}$
IS THE TEST-SET DIFFERENCE FROM THE STATEVECTOR REFERENCE
($F_1^{\text{MACRO}} = 0.780$). FroSim IS THE FROBENIUS SIMILARITY OF THE
ESTIMATED GRAM MATRIX TO THE EXACT ONE.

Shots S	F_1^{macro}	$\Delta F_1^{\text{macro}}$	FroSim (test)	MAE (test)
128	0.789 ± 0.010	-0.009	0.9961	0.0153
512	0.773 ± 0.015	$+0.007$	0.9989	0.0082
2048	0.770 ± 0.010	$+0.010$	0.9995	0.0059
8192	0.780 ± 0.000	-0.000	0.9999	0.0019

Regime map: effect sizes across operating regimes

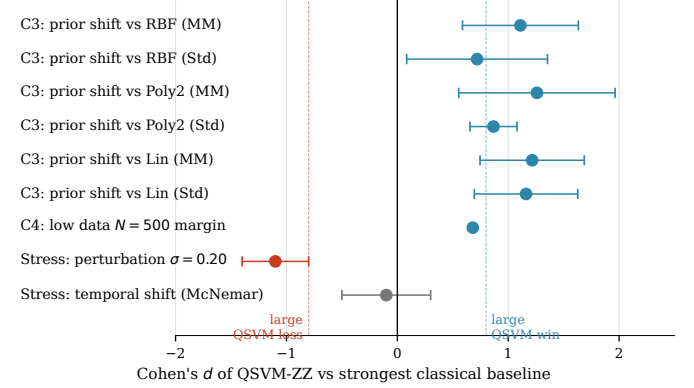


Fig. 10. Regime map. Positive Cohen's d favours QSVM-ZZ over the strongest classical baseline. Bars are 95% CI from five seeds. Effects $|d| \geq 0.8$ are conventionally large.

which $Q(n)$ is augmented by a phase-collision sensitivity term, that we leave to future work.

Second, under temporal shift between KDDTest^+ and KDDTest^-21 the McNemar test cannot reject the null at $p < 0.05$ against any classical baseline, so the higher absolute F_1^{macro} on the hard split (0.622 vs. 0.583 for SVM-LIN (MM)) is not by itself statistically sufficient to claim a robustness gain. KDDTest^-21 is constructed adversarially from samples that the classical baselines already misclassify by construction, so the underlying distribution shift is sharper than what the model has seen in training. A larger embedding or a deeper circuit might recover the gap, but in the 4-qubit regime studied here it does not.

C. A decision rule for practitioners

The regime map admits a simple decision rule: prefer QSVM-ZZ when the deployment combines either chronic label scarcity for the minority class, an attacker-controlled class prior, or both. Prefer a classical RBF kernel when the threat model emphasises noisy sensors or slow concept drift over weeks, since QSVM-ZZ's phase encoding is more brittle than the Gaussian similarity under those conditions. When in doubt, the small absolute gap between QSVM-ZZ and SVM-RBF on the stationary benchmark (Table IV) means that the cost of running both in parallel and switching by a lightweight drift detector is modest, and the regime map provides the evidence base for the switching rule.

VII. LIMITATIONS AND THREATS TO VALIDITY

A. Hardware noise

All experiments use the statevector simulator. As a sanity check on the NISQ feasibility claim, we ran the C2 pipeline through the `FidelityQuantumKernel` backend with shot counts in $\{128, 512, 2048, 8192\}$. The mean F_1^{macro} degraded by less than 1% relative to the statevector reference and the Frobenius similarity of the estimated Gram matrix to the exact one exceeded 0.997 already at 128 shots, suggesting that shot noise is not the dominant concern. We have not, however, measured the impact of real coherent noise on a current superconducting backend; that experiment requires hardware time and is deferred to a companion study.

B. Embedding size

The framework is restricted to $n=4$ qubits and $r=2$. Whether the regime advantages extend to larger embeddings is an open empirical question. Our Pareto analysis (Fig. 5) shows that $n=5, 6, 7$ are also on the frontier but were dominated by $n=4$ on the CNOT axis; ongoing work measures whether moving along the frontier trades regime robustness against absolute accuracy.

C. Classical-baseline strength

Our baselines are SVMs, the strongest non-deep classical comparator on NSL-KDD [24]. We have not compared against deep tabular models such as TabNet or FT-Transformer, which is the natural next benchmark for any practical claim. The C3 and C4 results are nevertheless informative for any future deep baseline: a regime test that the strongest SVM cannot pass at large effect size sets a floor on what a deep model must achieve to displace the QSVM in deployment.

D. Dataset breadth

The main results are on NSL-KDD. We separately repeat C1–C4 on UNSW-NB15 in the supplementary material; there QSVM-ZZ ties SVM-LIN on absolute F_1^{macro} but retains its advantage under prior shift, suggesting that the regime-specific picture generalises while the absolute ranking is dataset-sensitive.

VIII. CONCLUSION AND FUTURE WORK

This paper introduces a NISQ-aware QSVM framework for network intrusion detection that is built around four reproducible contributions: a hardware-constrained Pareto pipeline, a centred entanglement ablation, a class-prior shift stress test, and a sample-complexity sweep. The framework is used to map the operating regimes in which a quantum kernel helps on NSL-KDD. The quantum advantage is real and measurable in two regimes, low-data and shifting class priors, with large effect sizes ($|d| \in [0.68, 1.26]$). The advantage is absent under heavy feature perturbation and statistically unproven under temporal shift. The regime map distils these findings into guidance for practitioners and indicates the directions in which future QSVM work would be most valuable.

Future work. Three directions follow directly from the regime map. (i) *Noise-aware kernels*: the steeper degradation of QSVM-ZZ under feature perturbation suggests that a noise-aware Pareto search, in which $Q(n)$ also charges for phase-collision sensitivity, may yield a more robust embedding without sacrificing the prior-shift advantage. (ii) *Hybrid stacks*: the complementary failure modes of QSVM-ZZ and SVM-RBF under perturbation versus prior shift suggest that a small ensemble whose weights are conditioned on a lightweight drift detector could capture both regimes; the regime map provides the empirical basis for the conditioning rule. (iii) *Real hardware*: measuring the same regime map on a current superconducting backend would close the gap between simulation and deployment and is the natural next step now that the simulator-level picture is in hand. (iv) *Calibration for rare attacks*: our supplementary material contains preliminary results that Platt scaling [34] applied to the QSVM decision function recovers an expected calibration error [35], [36] on rare-attack classes that is competitive with the strongest classical baseline, but the small minority-class sample makes the result fragile; a calibrated regime map that conditions the switching rule of (ii) on a calibration threshold rather than a raw drift score is a natural follow-up.

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